

ABSTRACT

In the modern education system, students rely heavily on digital notes for learning and exam preparation. However, identifying complex and difficult topics from large volumes of notes is time-consuming and inefficient using traditional reading methods. To overcome this challenge, this project presents an **Offline AI-Based Notes Complexity Analyzer with Heatmap Visualization**.

The proposed system operates in an offline environment and uses **Natural Language Processing (NLP)** techniques to analyze textual notes provided by the user. The notes are segmented into meaningful sections and evaluated using parameters such as sentence length, presence of technical terminology, and textual structure. Based on these parameters, a complexity score is calculated for each section to determine its difficulty level.

The calculated complexity scores are visually represented using a **heatmap**, where color variations indicate easy, medium, and hard sections of the notes. This visual representation helps students quickly identify challenging topics and prioritize their study efforts more effectively. Since the system works completely offline, it ensures data privacy and usability in areas with limited or no internet connectivity.

The proposed solution is simple, efficient, and scalable, making it suitable for academic use. It can be further enhanced to support PDF documents, handwritten notes using OCR, and personalized learning recommendations. This project demonstrates the effective integration of artificial intelligence and data visualization techniques to improve the learning experience.